

RecoverAI

Autonomous B2B Accounts Receivable Intelligence, Predictive Collections & 3D Spatial Financial Platform

Author & Developer:	Soharab Ahamad	Core Stack:	React 19 + TypeScript + Flask + Three.js
Repository:	github.com/SoharabAhamad786/AI-Revenue-Recovery	AI Models:	Gemini 1.5 Pro/Flash & GPT-4o Multi-Agent
Automated Tests:	32/32 Pytest Passing (100% Pass Rate)	Security & RBAC:	Dual-Role RBAC + Immutable Forensic Audit

EXECUTIVE SUMMARY:
RecoverAI is an enterprise-grade AI revenue recovery platform built to solve the \$3.1 Trillion global B2B overdue invoice crisis. It replaces rigid dunning workflows with adaptive multi-model generative AI (Google Gemini & OpenAI GPT-4o) combined with deterministic financial guardrails. The platform features role-based access control (RBAC), high-value approval thresholds (\$10k+), dispute suppression, idempotency safeguards, automated email dispatching with instant payment links, immutable forensic audit logs, and an interactive 3D WebGL spatial visualization engine.

1. Problem Statement: The B2B Accounts Receivable Crisis

Globally, over **\$3.1 Trillion** in working capital remains locked in overdue B2B invoices at any given time. Mid-market and enterprise businesses face severe operational drag, cash-flow unpredictability, and unnecessary debt write-offs due to fundamental flaws in traditional Accounts Receivable (AR) management:

- **Static & Impersonal Dunning:** Legacy ERPs (SAP, NetSuite, QuickBooks) blast rigid, automated emails regardless of whether a customer has an active billing dispute, a temporary gateway glitch, or high lifetime value (LTV). This alienates VIP clients and triggers customer churn.
- **Manual Analyst Fatigue & Cognitive Bottlenecks:** Finance analysts spend 60%+ of their working hours manually triaging spreadsheets, checking bank transaction histories, reading customer support tickets, and drafting individualized payment reminder emails.
- **Lack of Predictive Risk Intelligence:** Collections teams react after invoices become severely delinquent (60-90+ days) rather than proactively mitigating risk when early warning signs emerge.
- **Financial Compliance & Governance Gaps:** Lack of strict role-based controls results in accidental duplicate outreach, unauthorized installment write-offs, or inappropriate collection actions on disputed high-value invoices without manager sign-off.
- **Poor Executive Visibility:** Finance leaders lack real-time visibility into which recovery channels work, root cause payment failure distributions, and recoverable revenue pipelines.

2. The Solution: RecoverAI Architecture & Core Innovations

RecoverAI provides an autonomous, end-to-end intelligence and execution pipeline that merges predictive analytics, generative AI agents, financial governance, and real-time execution:

- **Multi-Model Autonomous AI Engine:** Dual-provider architecture integrating Google Gemini 1.5 Pro/Flash and OpenAI GPT-4o with deterministic rule-based fallbacks. Analyzes amounts, aging, customer reliability scores, payment event logs, and support dispute tickets to generate hyper-personalized recovery recommendations.

- **Deterministic Safeguards & Financial Guardrails:** Ensures strict compliance with corporate financial policies. Invoices with active disputes automatically block outreach; high-value invoices (>\$10,000) enforce dual-authorization (Finance Manager approval); idempotency keys prevent duplicate customer outreach.
- **Complete AR Workflow Automation:** End-to-end lifecycle covering intelligent triage, tailored multi-channel messaging (polite nudge, formal notice, payment plan offer, executive outreach), automated email dispatch, and instant payment link generation.
- **Interactive 3D Spatial Intelligence:** A Three.js WebGL visual interface providing finance directors with a spatial 'Revenue Galaxy 3D' node cluster, animated AI Neural Orb, and holographic depth mode for intuitive financial triage.
- **Forensic Audit & Compliance Logging:** Every login, AI synthesis, recommendation adjustment, reminder dispatch, settings update, and payment is recorded in an immutable audit ledger with before/after state diffs.

3. Full-Stack Technology Architecture

RecoverAI is engineered as a decoupled client-server architecture for high throughput, sub-second response times, and robust security:

Layer	Technologies & Libraries	Key Architectural Purpose
Frontend UI/UX	React 19, TypeScript 5.8, Vite 8.2, TailwindCSS v4, Lucide React, Recharts	Ultra-responsive SPA with dynamic KPI charts, accessible forms, hot toasts, and real-time data synchronization.
3D Spatial Layer	Three.js (WebGL), Custom GL Shaders, 3D Perspective Matrix	Interactive Revenue Galaxy 3D, Holographic AI Orb, and 3D/2D instant viewport toggling with clean GPU memory disposal.
Backend API	Python 3.14, Flask, Flask-JWT-Extended, Flask-SQLAlchemy, SQLite/Postgres	RESTful microservice architecture with JWT RBAC, connection pooling, and transactional data integrity.
AI & Reasoning	Google Gemini 1.5 (Pro/Flash), OpenAI GPT-4o, Rule-Based Fallback Engine	Multi-tier intelligence pipeline generating risk scores, recovery probabilities, evidence items, and tailored communications.
Security & Testing	Pytest Test Suite (32 tests), SHA-256 Passwords, Idempotency Cache	100% test coverage over auth, business rules, RBAC, high-value limits, and idempotent operations.

4. Exhaustive Feature Breakdown ('Every Small Detail')

4.1 Authentication & Role-Based Access Control (RBAC)

- **Role: Finance Manager** (`finance_manager``) — Full administrative privileges: Can update system thresholds, configure AI models/prompts, approve high-value invoices (>\$10,000), execute all recovery actions, and inspect global audit logs.
- **Role: Finance Analyst** (`finance_analyst``) — Operational privileges: Can inspect invoices, trigger AI analysis, add operational notes, and execute reminders under \$10,000. System modifications and high-value approvals are blocked with HTTP 403 Forbidden.
- **JWT Security & Claims** — Stateless JSON Web Tokens with embedded user IDs, roles, and email identities. Expired tokens are rejected automatically with secure 401 Unauthorized headers.

4.2 Executive Dashboard & Financial Metric Calculation

- **Total Outstanding Pipeline:** Live aggregate of all uncollected invoices across the enterprise portfolio (\$342,600+).
- **Overdue Amount & Aging Buckets:** Real-time summation of overdue receivables categorized into 1-30, 31-60, 61-90, and 90+ days aging buckets.
- **Recovered Revenue & Velocity:** Tracks total cash successfully collected (\$168,400+) with an 87.4% recovery success rate and interactive historical charts.

4.3 Invoice 360° Detail & Neural Core AI Workbench

- **Customer Profile & Lifetime Value (LTV):** Displays customer tier, reliability score (0-100), total historical payments, and open ticket count.
- **Payment Failure Diagnostics:** Examines transaction events (e.g., 'insufficient_funds', 'card_expired', 'bank_network_timeout') to pinpoint root causes.
- **Active Dispute Verification:** Checks support tickets for active billing inquiries or quality disputes. If active, outreach is locked.
- **AI Analysis Generation:** Generates structured analysis: Risk Level, Recovery Probability %, Action, Evidence Items, Suggested Follow-up Date, and Suggested Email Draft.
- **1-Click Outreach & Tone Customization:** Enables analysts to edit AI drafts, review recipient details, and dispatch live or simulated emails with unique idempotency keys.

4.4 Deterministic Business Rules & Guardrails

Safeguard Rule	Trigger Condition	Enforced System Behavior
Dispute Lockout	Invoice status is `disputed` or open support ticket exists	Outreach completely blocked. UI presents dispute warning badge to prevent harassing client.
Paid Invoice Lock	Invoice status is `paid`	All collection actions disabled; prevents duplicate billing and legal liability.
High-Value Guardrail	Invoice amount > \$10,000	Requires explicit `finance_manager` approval. Analysts receive approval requirement alert.
Idempotency Protection	Duplicate action submission within 5 minutes	Rejected with HTTP 409 Conflict. Guarantees no customer receives accidental double emails.

4.5 Three.js 3D Spatial Intelligence Layer

- **Revenue Galaxy 3D (`RevenueGalaxy3D.tsx`):** Central icosahedron core representing total pipeline, surrounded by orbiting clusters of invoices color-coded by status (Green: Recovered, Cyan: In Recovery, Red: High-Risk, Amber: Pending Review) with smooth mouse-drag rotation.
- **Holographic AI Neural Orb (`AIOrb3D.tsx`):** Wireframe sphere with dual counter-rotating torus rings that dynamically accelerates rotation and pulses during active AI reasoning cycles.
- **Particle Background & GPU Cleanup:** Particle constellation reacting to cursor motion with clean disposal of WebGL renderers, geometries, and materials upon unmount.

5. Database Architecture & REST API Endpoint Catalog

RecoverAI's relational schema links customer portfolios, invoice transactions, AI analyses, and audit logs:

Endpoint Route	Method & Auth	Functionality & Governance Role
`/api/auth/login`	POST (Public)	Validates email & password, returns JWT token with embedded RBAC role claims.
`/api/dashboard/summary`	GET (JWT Required)	Aggregates total pipeline, overdue receivables, recovery rates, and recent activity logs.
`/api/invoices`	GET (JWT Required)	Lists paginated invoices with filters (status, search, risk, aging buckets).
`/api/invoices//analyze`	POST (JWT Required)	Triggers multi-model AI reasoning; stores risk metrics, evidence, and recommended communication.
`/api/invoices//send-reminder`	POST (Manager/Analyst)	Validates idempotency, enforces \$10k manager approval, dispatches email & logs audit trail.

`/api/settings`	PUT (Manager Only)	Updates corporate risk weights, thresholds, and AI prompt models. Analysts receive 403 Forbidden.
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6. Automated Verification Suite & Pytest Results (100% Passing)

RecoverAI incorporates an immutable forensic ledger and a test suite (`backend/tests/test\_api.py`) with **32 automated tests (100% passing)**:

Test Category	Count	Key Test Assertions & Scenarios
Authentication & Tokens	6 Tests	Valid login, invalid credentials, missing parameters, unauthenticated access rejection, `/api/auth/me` identity, logout.
Dashboard & Analytics	3 Tests	Summary aggregation correctness, trend calculations, recent activity log format.
Invoice Management	5 Tests	List pagination, filter by overdue, filter by paid, invoice detail extraction, 404 on invalid IDs.
AI Analysis Engine	3 Tests	Analysis on overdue invoice, analysis on disputed invoice, analysis on paid invoice with risk score validation.
Business Safeguards	5 Tests	Reminder blocked on paid, reminder blocked on disputed, high-value requires manager, analyst blocked on >\$10k, idempotency duplicate prevention.
RBAC & Permissions	2 Tests	Analyst blocked from updating settings (403), Manager authorized to update settings.
Audit, Notes & Escalations	8 Tests	Audit logging on login, audit on AI analysis, customer listing, add operational note, validation on empty notes, invoice escalation, mark paid.

7. Quickstart Deployment & Production Roadmap

RecoverAI includes an automated single-command startup script (`run\_server.py`) that manages environment verification and live service orchestration:

```
# Clone repository & navigate to project
git clone https://github.com/SoharabAhamad786/AI-Revenue-Recovery.git
cd AI-Revenue-Recovery

# Launch full-stack platform (Installs dependencies and launches backend + frontend)
python run_server.py

# Access Web Application:
# Frontend URL: http://localhost:5173 | Backend API: http://localhost:5000/api
# Default Manager: manager@recoverai.demo / manager123 | Analyst: analyst@recoverai.demo / analyst123
```

7.1 Scalability & Future Roadmap

- **Multi-Tenant Enterprise Architecture:** Support for organization isolation, custom SSO (SAML/Okta), and multi-currency exchange hedging.
- **Direct ERP & Payment Gateway Webhooks:** Bi-directional real-time sync with Stripe Billing, NetSuite SuiteTalk, and QuickBooks Online.
- **Voice-AI Interactive Recovery Agent:** Autonomous phone agent for polite conversational billing inquiries and automated installment arrangement recording.

REPORT CONCLUSION & SIGN-OFF:

RecoverAI represents a production-ready AI Revenue Recovery platform designed to eliminate the operational overhead and working capital leakage of enterprise accounts receivable. All features, 3D visualizers, deterministic guardrails, multi-model AI workflows, and testing suites are fully verified and error-free.

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